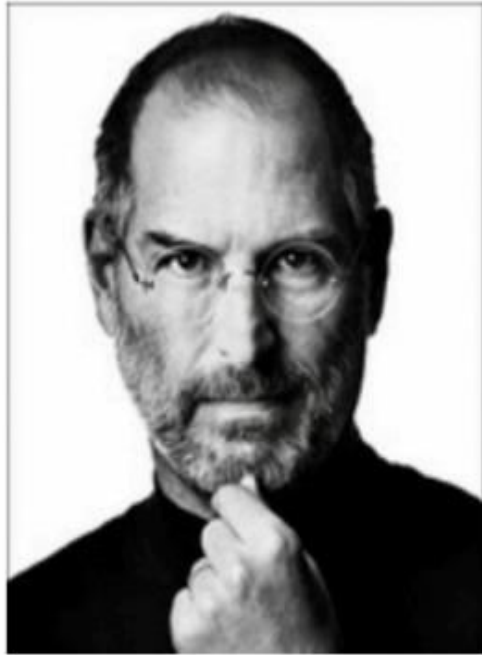




Software Engineering

LECTURE 6: REQUIREMENT ENGINEERING

What do we exactly want to build?"



— Steve Jobs

“Some people say, "Give the customers what they want." But that's not my approach. Our job is to figure out what they're going to want before they do. I think Henry Ford once said, "If I'd asked customers what they wanted, they would have told me, 'A faster horse!'" People don't know what they want until you show it to them. That's why I never rely on market research. Our task is to read things that are not yet on the page.”

Topics covered

- Introduction to Requirements engineering

- Requirement Engineering Process

- Elicitation
- Analysis
- Validation
- Evolution

- How to write good requirements

- Requirement's specification document

Requirements Engineering

- ❑ The process of establishing the services that the customer requires from a system and the constraints under which it operates and is developed
- ❑ The requirements themselves are the descriptions of the system services and constraints that are generated during the requirements engineering process

Not a phase
or stage!

Communication
is as important
as the analysis

Quality means
fitness-for-purpose.
Cannot say anything
about quality unless
you understand the
purpose

Requirements Engineering (RE) is a set of activities concerned with identifying and communicating the purpose of a software-intensive system, and the contexts in which it will be used. Hence, RE acts as the bridge between the real world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software-intensive technologies

Designers need to
know how and where
the system will be
used

Requirements are
partly about what
is needed...

...and partly about
what is possible

Need to identify all the stakeholders -
not just the customer and user

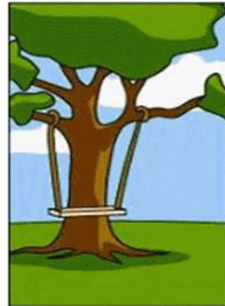
Definition of Requirement Engineering

Why we need
RE?

Need of Requirement Engineering



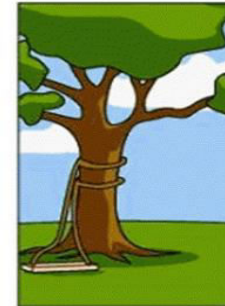
How the customer explained it



How the project leader understood it



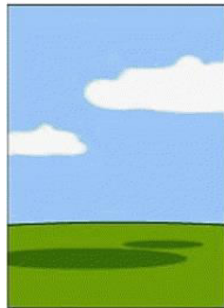
How the engineer designed it



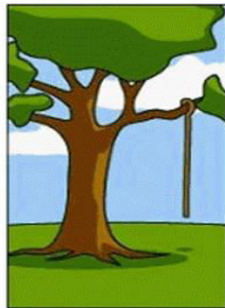
How the programmer wrote it



How the sales executive described it



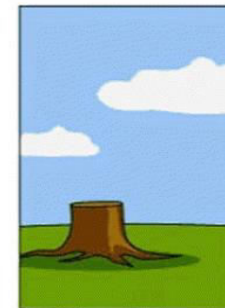
How the project was documented



What operations installed



How the customer was billed



How the help desk supported it



What the customer really needed

Types of requirement

User requirements

- Statements in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers.

System requirements

- A structured document setting out detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented so may be part of a contract between client and contractor.

User requirement definition

1. The MHC-PMS shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.

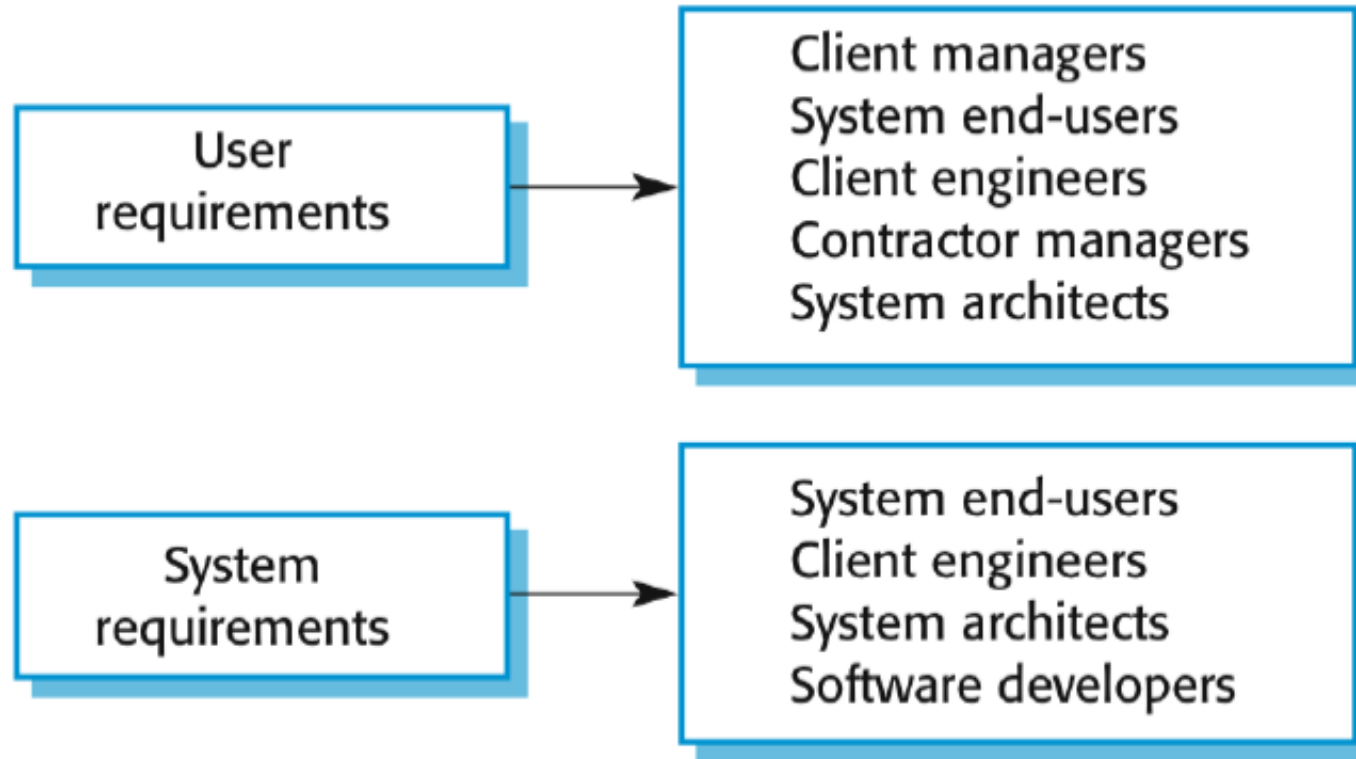
1.2 The system shall automatically generate the report for printing after 17.30 on the last working day of the month.

1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.

1.4 If drugs are available in different dose units (e.g. 10mg, 20 mg, etc.) separate reports shall be created for each dose unit.

1.5 Access to all cost reports shall be restricted to authorized users listed on a management access control list.

User and system requirements



Readers of
different
types of
requirements
specification

System stakeholders

Any person or organization who is affected by the system in some way and so who has a legitimate interest

Stakeholder types

- End users
- System managers
- System owners
- External stakeholders

Functional and nonfunctional requirements

☐ Functional requirements

- ☐ Statements of services the system should provide, **how** the system should react to particular inputs and how the system should behave in particular situations
- ☐ May state what the system should not do

☐ Non functional requirements

- ☐ Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
- ☐ Often apply to the system as a whole rather than individual features or services

☐ Domain requirements

- ☐ Constraints on the system from the domain of operation

Functional requirements

- Describe functionality or system services
- Depend on the type of software, expected users and the type of system where the software is used
- Functional user requirements may be high level statements of what the system should do
- Functional system requirements should describe the system services in detail

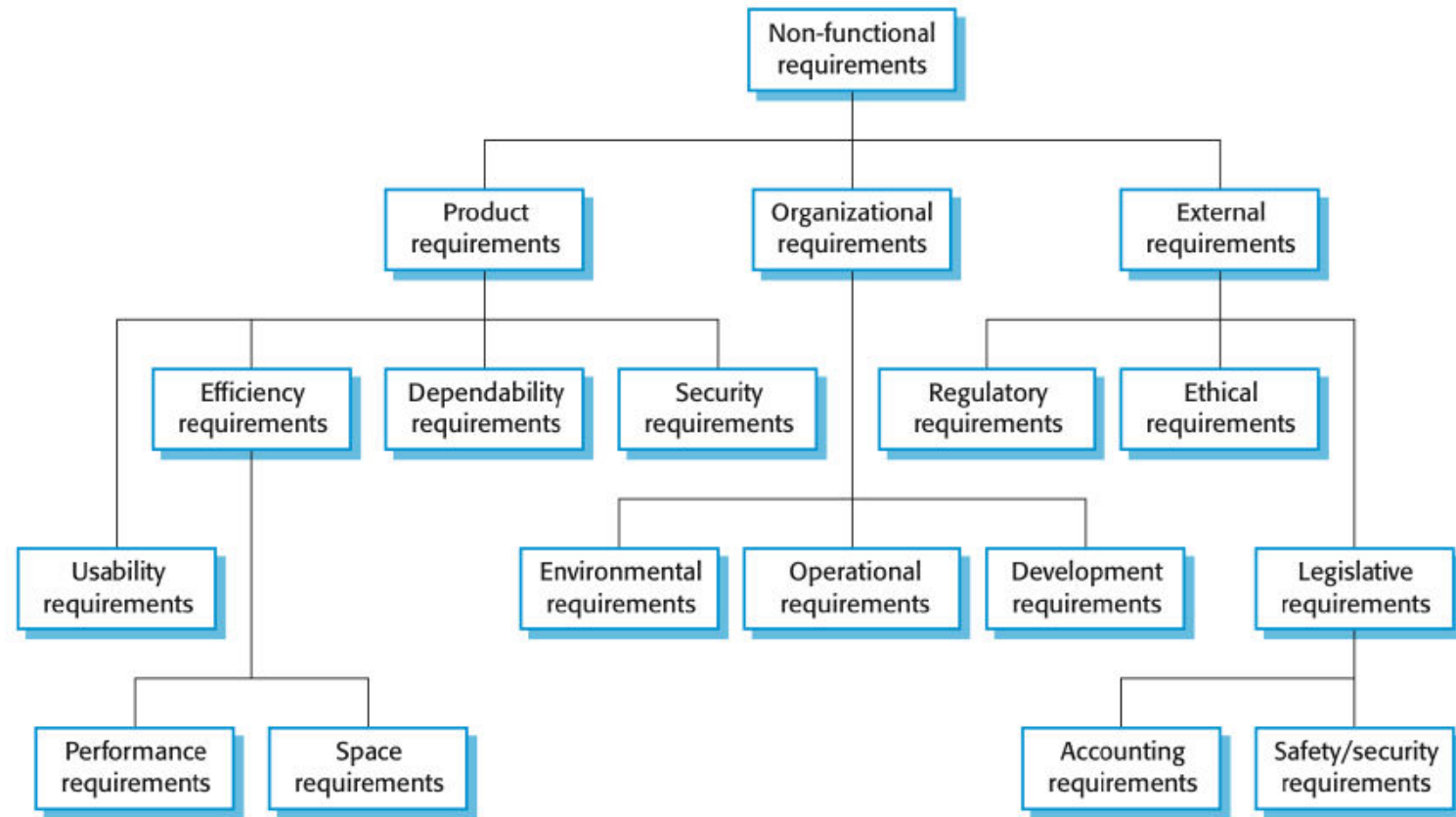
Functional requirements for the MHC PMS

- A user shall be able to **search** the appointments lists for all clinics
- The system shall **generate** each day, for each clinic, a list of patients who are expected to attend appointments that day
- Each staff member using the system shall be uniquely **identified** by his or her 8 digit employee number

Non functional requirements

- These define **system properties** and **constraints** e.g. reliability, response time, and storage requirements. Constraints are **I/O device capability**, **system representations**, etc.
- **Process requirements** may also be specified mandating a particular IDE, programming language or development method.
- Non functional requirements may be **more critical** than functional requirements. **If these are not met, the system may be useless.**

Types of nonfunctional requirement



Non functional requirements implementation

- Non functional requirements may affect the overall architecture of a system rather than the individual components
- For example, to ensure that performance requirements are met, you may have to organize the system to minimize communications between components
- A single non functional requirement, such as a security requirement, may generate a number of related functional requirements that define system services that are required
- It may also generate requirements that restrict existing requirements

Non functional classifications

Product requirements

- Requirements which specify that the **delivered product must behave** in a particular way e.g. execution speed, reliability, etc.

Organizational requirements

- Requirements which are a consequence of **organizational policies** and **procedures** e.g. process standards used, implementation requirements, etc.

External requirements

- Requirements which arise from **factors which are external** to the system and its **development process**, e.g. interoperability requirements, legislative requirements, etc.

Examples of nonfunctional requirements in the MHC PMS

- **Product requirement**

- The MHC PMS shall be available to all clinics during normal working hours (Mon Fri, 08.30 - 17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

- **Organizational requirement**

- Users of the MHC PMS system shall authenticate themselves using their health authority identity card.

- **External requirement**

- The system shall implement patient privacy provisions as set out in HStan 03 2006 priv.

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

Metrics for specifying nonfunctional requirements



Requirements engineering processes

Requirements engineering processes

- ✧ The processes used for RE vary widely depending on the **application domain**, the **people involved** and the **organization developing the requirements**
- ✧ However, there are a number of generic activities common to all processes
 - Requirements elicitation
 - Requirements analysis
 - Requirements validation
 - Requirements management
- ✧ In practice, RE is an iterative activity in which these processes are interleaved

Requirements elicitation and analysis

- ✧ Sometimes called *requirements elicitation* or *requirements discovery*
 - ✧ Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints
 - ✧ May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called *stakeholders*.
- 

Problems of requirements analysis

- ✧ Stakeholders don't know what they really want
- ✧ Stakeholders express requirements in their own terms
- ✧ Different stakeholders may have conflicting requirements
- ✧ Organizational and political factors may influence the system requirements
- ✧ The requirements change during the analysis process. New stakeholders may emerge and the business environment may change

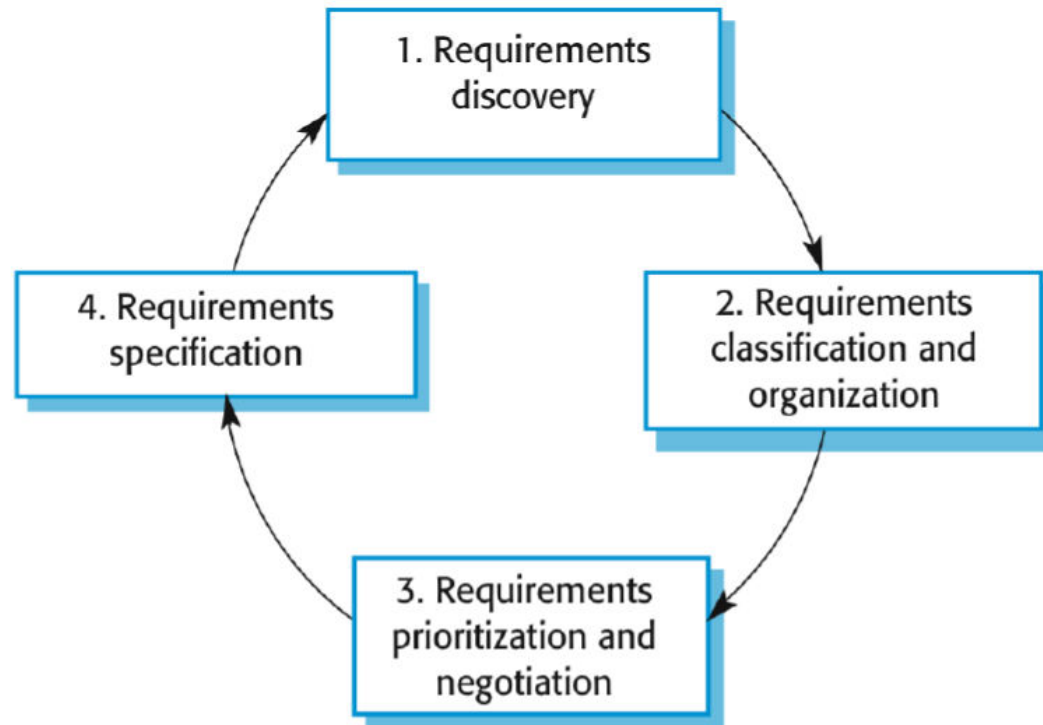
Requirements elicitation and analysis

✧ Software engineers work with a range of system stakeholders to find out about the application domain, the **services that the system should provide**, the **required system performance**, hardware constraints, other systems, etc.

✧ Stages include:

- Requirements **discovery**
- Requirements **classification and organization**
- Requirements **prioritization and negotiation**
- Requirements **specification**

The requirements elicitation and analysis process



Process activities

✧ Requirements discovery

- **Interacting with stakeholders** to discover their requirements. **Domain requirements** are also discovered at this stage

✧ Requirements classification and organization

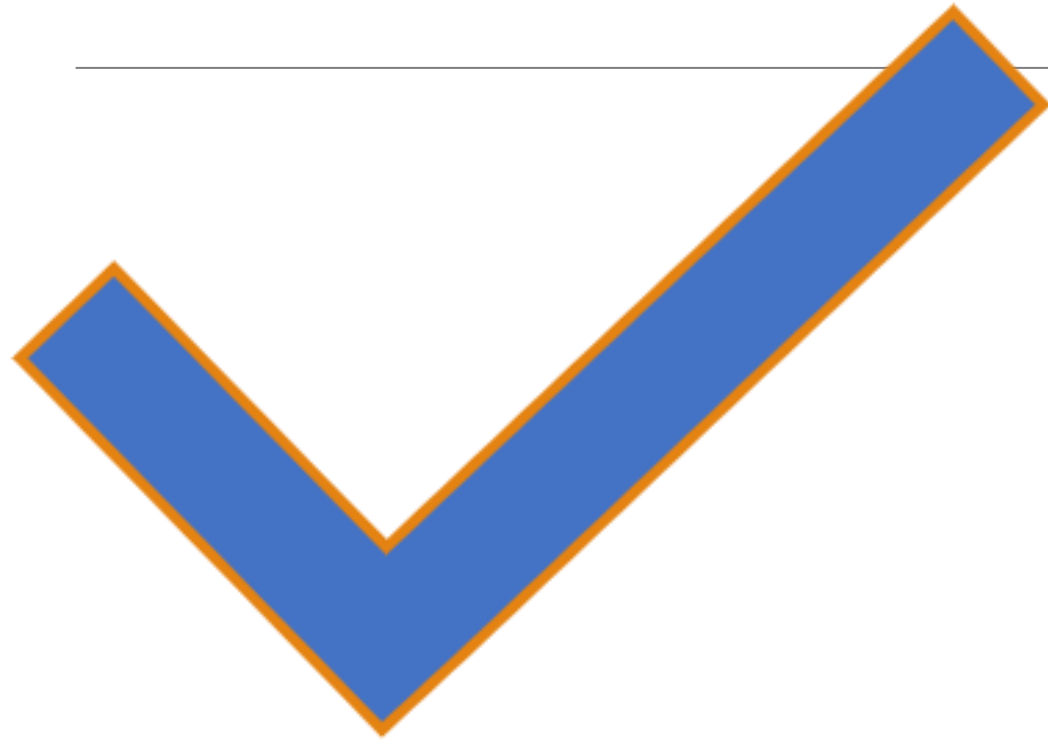
- **Groups related requirements** and **organizes** them into coherent clusters

✧ Prioritization and negotiation

- Prioritizing requirements and resolving requirements conflicts

✧ Requirements specification

- Requirements are documented and input into the next round of the spiral



REQUIREMENT S ELICITATION

Elicitation Techniques

- Brainstorming
- Document Analysis
- Focus Groups
- Interface Analysis
- Interviews
- Observation
- Prototyping
- Requirements Workshops
- Survey/Questionnaire

Requirements Elicitation techs

Traditional techniques questionnaires and surveys, interviews, and analysis of existing documentation such as organisational charts, process models or standards, and user or other manuals of existing systems.

Group elicitation techniques They include brainstorming and focus groups, as well as RAD/JAD workshops (using consensus-building workshops with an unbiased facilitator)

Contextual techniques ethnographic techniques such as participant observation.

Requirements Elicitation techs...

Cognitive techniques *protocol analysis* (in which **an expert thinks aloud** while performing a task, to provide the observer with insights into the cognitive processes used to perform the task),

laddering (using **probes** to elicit structure and content of stakeholder knowledge),

card sorting (asking stakeholders to **sort cards in groups**, each of which has name of some domain entity),

Laddering

Interviewer: What do you understand by 'EPR user training'?

Nurse: That the teacher is able to teach me the most basic functionalities in the system.

Interviewer: What do you understand by the most basic functionalities in the system?

Nurse: How to write a nursing note, save it and be able to find it again. But these functionalities also concern procedures regarding the overall hospitalization of the patient.

Interviewer: Why is it important that the teacher is able to teach you these functionalities in the system?

Nurse: Because it makes me confident when I have to use the system later on. Then I know that I will be able to carry out my work as a nurse.

Interviewer: What do you mean by 'feeling confident' and why is it important that you feel confident when using the system?

Nurse: Well, I wish to provide the best possible patient care...

Card Sorting



Ethnography

- ✧ A **social scientist** spends a considerable time observing and analyzing how people actually work
- ✧ People do not have to explain or articulate their work
- ✧ **Social and organizational factors** of importance may be observed
- ✧ Ethnographic studies have shown that work is usually **richer** and **more complex** than suggested by simple system models

Scope of ethnography

- ✧ Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work
- ✧ Requirements that are derived from **cooperation** and **awareness** of other people's activities
 - Awareness of what other people are doing leads to changes in the ways in which we do things
- ✧ Ethnography is effective for understanding existing processes but **cannot identify new features** that should be added to a system

Focused ethnography

- ✧ Developed in a project studying the air traffic control process
- ✧ Combines ethnography with prototyping
- ✧ Prototype development results in unanswered questions which focus the ethnographic analysis
- ✧ The **problem** with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant

Requirements validation

- ✧ Concerned with demonstrating that the requirements define the system that the customer really wants
- ✧ Requirements error costs are high so validation is very important
 - Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error

Requirements checking

- ✧ **Validity**. Does the system provide the functions which best support the customer's needs?
- ✧ **Consistency**. Are there any requirements conflicts?
- ✧ **Completeness**. Are all functions required by the customer included?
- ✧ **Realism**. Can the requirements be implemented given available budget and technology
- ✧ **Verifiability**. Can the requirements be checked?

Requirements validation techniques

✧ Requirements reviews

- Systematic manual analysis of the requirements

✧ Prototyping

- Using an executable model of the system to check requirements.

✧ Test-case generation

- Developing tests for requirements to check testability

Requirements reviews

- ✧ Regular reviews should be held while the requirements definition is being formulated
- ✧ Both client and contractor staff should be involved in reviews
- ✧ Reviews may be **formal** (with **completed documents**) or **informal**. **Good communications** between developers, customers and users can resolve problems at an early stage.

Review checks

✧ Verifiability

- Is the requirement realistically testable?

✧ Comprehensibility

- Is the requirement properly understood?

✧ Traceability

- Is the origin of the requirement clearly stated?

✧ Adaptability

- Can the requirement be changed without a large impact on other requirements?

Requirements Prioritization using Weigers Prioritization technique

Allows you to prioritize requirements based on benefits, detriment and risk involved

Helps in resolving conflicts and prioritize requirements

Relative Weight	2	0.5			1		1			
Requirements	Relative Benefit	Relative Detriment	Value Total	Value%	Relative Cost	Cost%	Relative Risk	Risk%	Priority	Rank
R1	6	3			2		1			
R2	9	7			4		2			
R3	4	7			4		2			
R4	2	1			1		1			
R5	4	9			4		4			
R6	4	8			3		3			
Total	29	35			15		11			

Weigers Prioritization technique Steps

Step 1: Find Value total

Value Total = Relative Benefit x Weight + Relative Determent x Weight

Value Total = Benefit* Relative Benefit Weight + Determent* ReltiveDetermentWeight

Step 2: Sum Value Total

Step 3: Find Value%

$$\text{Value \%} = \frac{\text{Value total (Requirement)}}{\text{Sum value total}} \times 100$$

Step 4: Find Cost%

$$\text{Cost \%} = \frac{\text{Cost of Requirement}}{\text{Total Cost}} \times 100$$

Step 5: Find Risk%

$$\text{Risk \%} = \frac{\text{Relative Risk of Requirement}}{\text{Total Relative Risk}} \times 100$$

Relative Weight	2	0.5			1		1			
Requirements	Relative Benefit	Relative Detriment	Value Total	Value%	Relative Cost	Cost%	Relative Risk	Risk%	Priority	Rank
R1	6	3	13.5	17.88079	2	11.11111	1	7.692308		
R2	9	7	21.5	28.47682	4	22.22222	2	15.38462		
R3	4	7	11.5	15.23179	4	22.22222	2	15.38462		
R4	2	1	4.5	5.960265	1	5.555556	1	7.692308		
R5	4	9	12.5	16.55629	4	22.22222	4	30.76923		
R6	4	8	12	15.89404	3	16.66667	3	23.07692		
Total	29	35	75.5		18		13			

Weigers Prioritization technique Steps

Step 6: Find Priority Score

$$\text{Priority} = \frac{(\text{Value\%})}{\text{Cost\%} * \text{Relative Cost Weight} + \text{Risk\%} * \text{Relative Risk Weight}}$$

$$\text{Priority} = \frac{\text{Benefit} * \text{Relative Benefit Weight} + \text{Determent} * \text{Relative Determent Weight}}{\text{Cost\%} * \text{Relative Cost Weight} + \text{Risk\%} * \text{Relative Risk Weight}}$$

If all weights = 1 then

$$\text{Priority} = \frac{\text{Benefit} + \text{Determent}}{\text{Cost} + \text{Risk}} = \frac{B + D}{C + R}$$

Step 7: Rank

Based on the priority, rank the requirements where 1 demotes the highest priority

Weigers Prioritization technique

Solution

Relative Weight	2	0.5			1		1			
R1	6	3	13.5	17.88079	2	11.11111	1	7.692308	0.950933	1
R2	9	7	21.5	28.47682	4	22.22222	2	15.38462	0.757225	2
R3	4	7	11.5	15.23179	4	22.22222	2	15.38462	0.405027	4
R4	2	1	4.5	5.960265	1	5.555556	1	7.692308	0.449904	3
R5	4	9	12.5	16.55629	4	22.22222	4	30.76923	0.312433	6
R6	4	8	12	15.89404	3	16.66667	3	23.07692	0.399915	5
Total	29	35	75.5		18		13			

Requirements management

- ✧ *Requirements management* is the process of **managing changing requirements** during the requirements engineering process and system development
- ✧ New requirements emerge as a system is being developed and after it has gone into use
- ✧ You need to keep **track of individual requirements** and **maintain links between dependent requirements** so that you can assess the impact of requirements changes. You need to establish a **formal process** for making change proposals and linking these to system requirements.

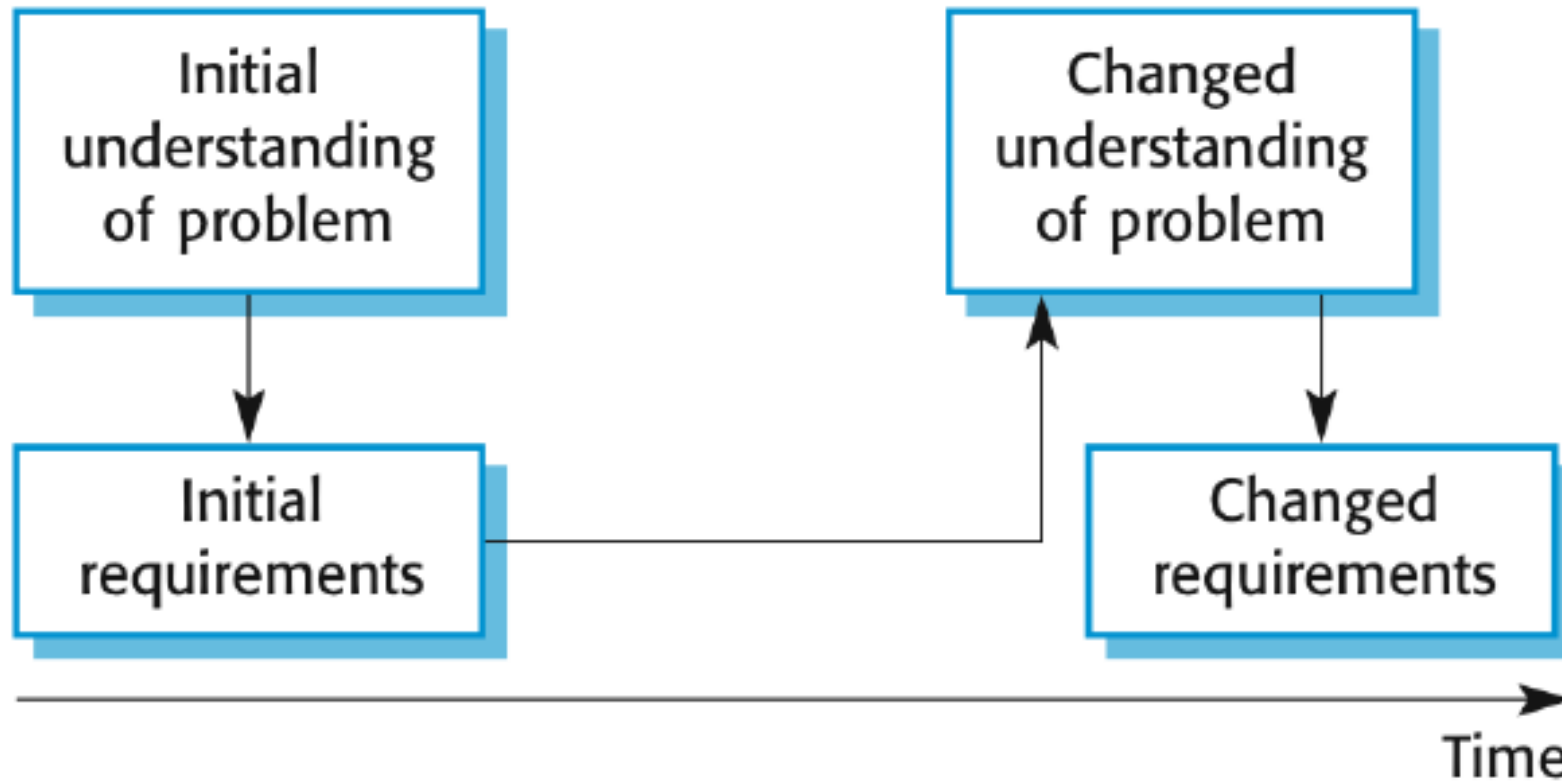
Changing requirements

- ✧ The **business and technical environment** of the system always changes after installation
 - New **hardware** may be introduced, it may be necessary to **interface the system with other systems**, **business priorities** may change (with consequent changes in the system support required), and **new legislation and regulations** may be introduced that the system must necessarily abide by
- ✧ The people who **pay for a system** and the **users of that system** are rarely the same people
 - System customers impose requirements because of **organizational** and **budgetary** constraints. These may conflict with end-user requirements and, after delivery, **new features** may have to be added for user support if the system is to meet its goals

Changing requirements

- ✧ Large systems usually have a **diverse user community**, with many users having **different requirements** and **priorities** that may be conflicting or contradictory
 - The final system requirements are inevitably a **compromise** between them and, with **experience**, it is often discovered that the balance of support given to different users has to be changed

Requirements evolution



Requirements management planning

- ✧ Establishes the **level of requirements management detail** that is required.
- ✧ Requirements management decisions:
 - **Requirements identification** Each requirement must be **uniquely identified** so that it can be cross-referenced with other requirements
 - **A change management process** This is the set of activities that assess the **impact and cost of changes**. I discuss this process in more detail in the following section.
 - **Traceability policies** These policies define the **relationships between each requirement** and **between the requirements and the system design** that should be recorded
 - **Tool support** Tools that may be used range from **specialist requirements management systems** to spreadsheets and simple database systems

Requirements change management

✧ Deciding if a requirements change should be accepted

- Problem analysis and change specification

- During this stage, the problem or the change proposal is analyzed to check that it is valid. This analysis is fed back to the change requestor who may respond with a more specific requirements change proposal, or decide to withdraw the request.

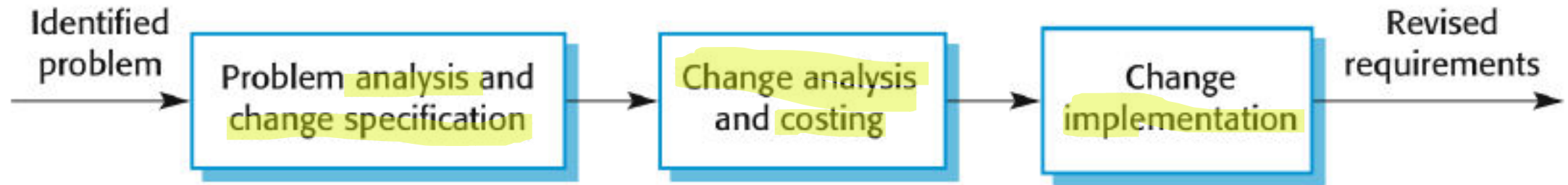
- Change analysis and costing

- The effect of the proposed change is assessed using traceability information and general knowledge of the system requirements. Once this analysis is completed, a decision is made whether or not to proceed with the requirements change.

- Change implementation

- The requirements document and, where necessary, the system design and implementation, are modified. Ideally, the document should be organized so that changes can be easily implemented.

Requirements change management



Key points

- ✧ Requirements for a software system set out what the system should do and define constraints on its operation and implementation
- ✧ **Functional requirements** are statements of the services that the system must provide or are descriptions of how some computations must be carried out
- ✧ **Non-functional requirements** often constrain the system being developed and the development process being used
- ✧ They often relate to the emergent properties of the system and therefore apply to the system as a whole

Key points

- ✧ The **software requirements** document is an agreed **statement of the system requirements**. It should be organized so that both system customers and software developers can use it.
- ✧ The **requirements engineering** process is an iterative process including requirements elicitation, specification and validation
- ✧ **Requirements elicitation and analysis** is an iterative process that can be represented as a spiral of activities –requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation

Key points

- ✧ You can use a range of **techniques for requirements elicitation** including interviews, use-cases and ethnography
- ✧ **Requirements validation** is the process of checking the requirements for validity, consistency, completeness, realism and verifiability
- ✧ **Business, organizational and technical changes** inevitably lead to changes to the requirements for a software system. Requirements management is the process of managing and controlling these changes.

